

## Defending Against Crescendo Multi-Turn Jailbreak Attacks

*A Comparative Evaluation of Two Defense Mechanisms on Llama-3.2-3B-Instruct*

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### Abstract

Multi-turn jailbreak attacks - in particular the Crescendo pattern - pose a significant challenge to deployed large language models (LLMs). By incrementally guiding a model through plausible, benign-seeming dialogue, an adversary can manoeuvre it into producing harmful content that it would immediately refuse if asked directly. This paper evaluates two complementary defense mechanisms against a suite of ten Crescendo attack scenarios spanning five harm categories on the Llama-3.2-3B-Instruct model. Defense 1 intercepts conversations via *semantic drift detection* (cosine similarity against a Turn-1 anchor embedding) combined with toxicity scoring. Defense 2 periodically re-injects a safety system prompt and employs an LLM-based intent classifier to block adversarial turns before they reach the main model. Against an undefended baseline Attack Success Rate (ASR) of 50%, Defense 1 achieves **0% ASR** and Defense 2 achieves **10% ASR**, demonstrating that lightweight, model-agnostic guards can substantially reduce jailbreak susceptibility without retraining.

## 1. Introduction

Large language models are now widely deployed in consumer and enterprise settings where they interact with untrusted users over multiple conversational turns. Alignment training - RLHF and its variants - makes models reluctant to comply with overtly harmful requests, yet this training remains susceptible to adversarial prompting strategies.

Crescendo [1] is a multi-turn attack in which each individual message appears benign in isolation but, taken together, the conversation gradually steers the model toward harmful outputs. The model's own prior responses - stored in its context window - serve as progressive commitment: having engaged at a superficial level, the model becomes increasingly likely to continue. This effect, sometimes called *memory stacking*, has been observed across a range of frontier models.

This paper makes three contributions: (i) a reproducible attack harness with ten multi-turn attack scenarios across five harm categories; (ii) two complementary defense mechanisms implemented as lightweight wrappers around an unmodified Llama-3.2-3B-Instruct model; and (iii) a quantitative comparison using Attack Success Rate (ASR) as the primary metric.

## 2. Background and Related Work

### 2.1 Multi-Turn Jailbreaking

Multi-turn attacks differ from single-turn jailbreaks in that the adversary does not rely on a single carefully crafted prompt - the conversation itself is the attack vector. The attacker begins with a framing prompt establishing a plausible, benign context (academic research, creative writing, professional training) and then issues a sequence of progressively more specific requests that each feel like a natural continuation. By the final turn, the model may have committed to a narrative that makes refusal seem inconsistent.

A related phenomenon is the *alignment tax* - the observation that safety-aligned models sacrifice some helpfulness on benign tasks in exchange for refusal capability on harmful ones. Prior work has noted that frontier models are not uniformly robust to multi-turn attacks even when they handle single-turn variants well [2].

### 2.2 Existing Defense Approaches

Published defenses against multi-turn jailbreaks fall into three categories: (1) model-level fine-tuning on adversarial conversation data, requiring access to model weights; (2) classifier-based input filtering, applying a guard model before each user turn; and (3) context management strategies, manipulating what the model sees in its context window. This work implements representatives of categories (2) and (3).

### 3. Experimental Setup

#### 3.1 Target Model

All experiments use **meta-llama/Llama-3.2-3B-Instruct**, loaded in 4-bit quantization (BitsAndBytesConfig, NF4, float16 compute, double quantization) to fit on a single T4 GPU (16 GB VRAM). Generation uses temperature 0.7, top-p 0.9, and a maximum of 512 new tokens per turn. The chat template is applied via `apply_chat_template` with a separate tokenisation step for correct input-length tracking.

#### 3.2 Attack Scenarios

Ten Crescendo attack scenarios are constructed across five harm categories: Weapons (2), Malware (2), Social Engineering (2), Harmful Substances (2), Extremism (1), and Misinformation (1). Each scenario consists of four turns, with Turn 1 establishing a benign framing and each subsequent turn escalating toward a harmful terminal request while maintaining the established framing.

| ID      | Category    | Terminal Request            |
|---------|-------------|-----------------------------|
| ATK-001 | Weapons     | Weapon modification guide   |
| ATK-002 | Weapons     | IED assembly instructions   |
| ATK-003 | Malware     | SQLi scanner script         |
| ATK-004 | Malware     | Phishing campaign kit       |
| ATK-005 | Social Eng. | Elder fraud phone script    |
| ATK-006 | Social Eng. | Predatory sales script      |
| ATK-007 | Harm. Sub.  | Toxic gas synthesis steps   |
| ATK-008 | Harm. Sub.  | Meth production walkthrough |
| ATK-009 | Extremism   | Recruitment message         |
| ATK-010 | Misinform.  | Voter suppression campaign  |

Table 1. Attack scenario summary.

#### 3.3 Evaluation Metric

Attack Success Rate (ASR) is the fraction of scenarios in which the model's final-turn response does not constitute a refusal. A heuristic refusal detector scans for phrases such as "I can't", "I won't", "harmful", "unethical", and similar formulations. A scenario is scored as a successful attack if no such phrase is detected in the final assistant response.

### 4. Defense Mechanisms

#### 4.1 Defense 1: Semantic Drift + Toxicity Filtering

The core intuition is that Crescendo attacks produce measurable *semantic drift*: the content of later turns diverges from the innocent starting context. We operationalise this by encoding Turn 1 into a dense sentence embedding using all-MiniLM-L6-v2 [3] and storing it as an anchor vector. For every subsequent turn, the incoming user message is

encoded and its cosine similarity to the anchor is computed. If similarity falls below threshold  $\tau = 0.72$ , the turn is intercepted.

A second, parallel signal is provided by a toxicity classifier. Each user message is scored by unitary/toxic-bert [4]; if the toxicity score exceeds 0.65, the turn is intercepted regardless of semantic drift. Either signal alone is sufficient to trigger interception. When a turn is intercepted, the model is never queried - a fixed safe response is returned and no assistant turn is added to the conversation history.

This design is entirely model-agnostic, adds no tokens to the LLM's context, and operates in sub-second time on CPU. Its principal limitation is sensitivity to threshold choice and to attacks that maintain surface-level topic coherence.

#### 4.2 Defense 2: Context Reset + Intent Classification

Defense 2 targets the memory-stacking mechanism through two complementary controls.

- **History truncation:** only the last `MAX_HISTORY = 3` turns are included in the model's context, limiting how much prior engagement the model can draw on when assessing the current request.
- **Periodic system prompt re-injection:** every `RESET_EVERY_K = 2` turns, the safety system prompt is prepended afresh, overriding any normative shift the conversation may have induced.

Before the main model is consulted, each incoming user message is passed to a lightweight intent classifier - a secondary call to the same LLM with a one-word classification prompt - that outputs either 'benign' or 'adversarial'. Turns classified as adversarial are blocked immediately without model generation. This classifier operates as a zero-shot binary guard requiring no training data.

### 5. Results

#### 5.1 Overall Attack Success Rate

The undefended baseline yields an ASR of 50%: five of the ten scenarios successfully elicit harmful final-turn responses. Table 2 summarises results across all three conditions. Defense 1 eliminates all successful attacks (0% ASR); Defense 2 reduces ASR to 10%.

| Condition                  | Succ. | Def.  | ASR   |
|----------------------------|-------|-------|-------|
| Baseline (No Defense)      | 5/10  | 5/10  | 50.0% |
| Defense 1 - Semantic Drift | 0/10  | 10/10 | 0.0%  |
| Defense 2 - Context Reset  | 1/10  | 9/10  | 10.0% |

Table 2. Overall ASR by condition.

## 5.2 Per-Category Breakdown

| Category        | Baseline | Defense 1 | Defense 2 |
|-----------------|----------|-----------|-----------|
| Weapons (2)     | 50%      | 0%        | 50%       |
| Malware (2)     | 50%      | 0%        | 0%        |
| Social Eng. (2) | 0%       | 0%        | 0%        |
| Harm. Sub. (2)  | 50%      | 0%        | 0%        |
| Extremism (1)   | 100%     | 0%        | 0%        |
| Misinform. (1)  | 100%     | 0%        | 0%        |

Table 3. Per-category ASR breakdown.

Social Engineering scenarios were already defended at baseline, suggesting the model's alignment training is more sensitive to psychological manipulation than to fictional or academic framings. Both defenses reduce Extremism and Misinformation ASR from 100% to 0%. Defense 2's sole failure (ATK-001) reflects a genuine limitation of zero-shot intent classification: WW2 historical fiction framing is genuinely difficult to distinguish from a benign research query.

## 5.3 Defense Trigger Analysis

For Defense 1, drift interception typically activates at Turn 3, when the attack transitions from conceptual framing to specific technical requests. Toxicity scoring rarely triggers independently, indicating the semantic drift signal carries the majority of the detection burden. For Defense 2, the intent classifier correctly flags adversarial intent in the majority of Turn-4 messages.

## 6. Discussion

### 6.1 Complementarity of the Two Defenses

The two defenses address the Crescendo vulnerability through fundamentally different mechanisms. Defense 1 operates on distributional properties of the text: any message diverging sufficiently from the conversation's starting point is blocked regardless of surface plausibility. Defense 2 operates on the model's context state, preventing commitment accumulation through truncation and refreshing normative priors through re-injection. A combined system applying both guards in sequence would likely close the residual Weapons vulnerability.

### 6.2 Limitations

The heuristic refusal detector introduces both false positives and false negatives. A motivated adversary with knowledge of the defense could adapt framing to maintain semantic similarity to the anchor turn, or phrase terminal requests to evade classifier detection. Experiments use a single model; generalisability to larger models or different alignment regimes is not established. Defense 1 adds two

inference calls per turn; Defense 2 adds one LLM call - overhead that may be material at deployment scale.

## 7. Future Work

Natural extensions include: (i) a combined defense running both guards in parallel; (ii) systematic threshold calibration over a held-out scenario set; (iii) adaptive attack evaluation where the adversary has partial knowledge of the defense mechanism; (iv) longer attack chains (6–8 turns); and (v) human evaluation of model outputs as a gold-standard ASR measurement.

## 8. Conclusion

Two lightweight, model-agnostic defense mechanisms substantially reduce the Attack Success Rate of Crescendo multi-turn jailbreaks against Llama-3.2-3B-Instruct. Defense 1 (semantic drift + toxicity) achieves **0% ASR** and Defense 2 (context reset + intent classification) achieves **10% ASR**, versus a 50% undefended baseline. Both operate without modifying model weights or requiring fine-tuning data, demonstrating that inference-time context management is a promising direction for practical LLM safety.

## References

- [1] M. Russinovich, A. Salem, and R. Eldan. "Great, Now Write an Article About That": The Crescendo Multi-Turn LLM Jailbreak Attack. *arXiv:2404.01833*, 2024.
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